

The uncontroversial 'thingness' of AI

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Big Data & Society
July–December: 1–5
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DOI: 10.1177/20539517231206794
journals.sagepub.com/home/bds



Abstract

This commentary starts with the question 'How is it that AI has come to be figured uncontroversially as a thing, however many controversies "it" may engender?' Addressing this question takes us to knowledge practices that philosopher of science Helen Verran has named a 'hardening of the categories', processes that not only characterise the **onto-epistemology of AI** but also are central to its constituent techniques and technologies. In a context where the stabilization of AI as a figure enables further investments in associated techniques and technologies, AI's status as controversial works to reiterate both its ontological status and its agency. It follows that interventions into the field of AI controversies that fail to trouble and destabilise the figure of AI risk contributing to its uncontroversial reproduction. This is not to deny the proliferating data and compute-intensive techniques and technologies that travel under the sign of AI but rather to call for a keener focus on their locations, politics, material-semiotic specificity, and effects, including their ongoing enactment as a singular and controversial object.

Keywords

Artificial intelligence critique, AI controversy, algorithmic practices, categorization, figuration, machine learning

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The goal of the question is to ferret out how relations and practices get mistaken for nontropic things-in-themselves in ways that matter to the chances for liveliness of humans and nonhumans. (Haraway, 1997: 141)

Across media, policy documents, and academic writings, statements regarding the ubiquity of AI are now a commonplace. Even those engaged in critical analysis frequently open with an affirmation of the proposition that AI, positioned as the active subject, is expanding in its presence and significance, a fact that motivates the urgency of a response. Treated as self-evident rather than in need of substantiation, this proposition constitutes the starting premise for whatever follows. In contrast, I want to propose that we treat the existence of AI itself as controversial. The point of doing so is not to deny the achievements and injuries of data-intensive algorithmic practices but rather to challenge the misplaced concreteness¹ that the nominalisation 'AI' effects. Put another way, **my argument is that the thingness of AI, its status as a stable and agential entity, needs to be made controversial: that we need to prioritize critical engagement with the work being done by**

the figure of AI in specific contexts. To let the term pass is to miss the opportunity to trace its sources of power and to demystify its referents.²

As the epigraph from Haraway suggests, critical scholarship requires attention to the rhetorical moves through which relations and practices are obscured in the naming of commodified things. For the purposes of this commentary the question is this: Just what are we talking about when we talk about 'AI'? The 'we' here refers both to those advancing prominent AI discourses and to our own writings as critical scholars. As critical scholars, our task is to challenge discourses that position AI as ahistorical, mystify 'its' agency and/or deploy the term as a floating signifier. Our task is also to be accountable to the question ourselves.

*the importance of contextual specificity
-this is an important aspect of developing a
critical AI literacy!*

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Fortunately, a growing body of critical scholarship provides resources for challenging dominant discourses and for the respecification and demystification of AI, widening the frame to include relevant genealogies, material practices and politics. If AI is presented as ahistorical – as a kind of *sui generis* technological agent – tracing the lineages of the field's unquestioned assumptions, including its presentism, is crucial. If AI is staged as a kind of mysterious or magical force, articulating the project's constituent materialities and technical practices, as well as the political economies that underwrite it, is an important ethicopolitical intervention. If AI is cited as if its referent were self-evident, asking what work that rhetorical stance is doing is a priority. In the next section, I offer indicative examples of each of these approaches to undoing the thingness of AI, as pathways to critical engagement and the formulation of counter-narratives.

AI as an historical subfield of computer and cognitive science

Critical genealogies of AI helpfully complicate origin stories that trace a linear progression from the emergence of machine models of mind in 17th century Europe to their formalization in mid-20th century cybernetics, cognitive science and computing.³ Histories of AI as a field typically locate its beginnings in the document that introduced the term, the Dartmouth Summer Research Project proposal 'to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it' (McCarthy et al., 1955). Examining the field's onto-epistemic legacy from a feminist standpoint, Adam (1998) emphasizes the founding fathers' reliance on key enabling premises that provide a through line across changes in techniques and technologies. These are a universalized figure of the knowing subject, simple realist assumptions about the significance of objects and erasures of the specificities of embodiment, location and relations in knowledge practices (see also Roberge and Castelle, 2021). Adam identifies AI's implicit knower as the canonical 'disinterested moral philosopher' (1998: 77), taken as the universal or interchangeable subject within a narrow membership group (composed historically of propertied, educated men). In contrast, she points out, feminist epistemology is concerned with the specificity of the knowing subject, the 'S' in propositional logics 'S knows that *p*'. As Adam observes, asking 'Who is *S*?' is not considered a proper concern for traditional epistemologists (1998: 77). She takes as exemplary cases Soar, the project to implement Allen Newell and Herbert Simon's conception of a general human problem solver in the early 1970s (Newell and Simon, 1972), and Cyc, the effort of Douglas Lenat and colleagues (Guha and Lenat, 1990) beginning in the 1980s to design and build an

encyclopaedic repository of 'human consensus knowledge' that could serve as a foundation for more robust and flexible expert systems. While Soar posited 'problem-solving' as a general, domain-independent procedure, Cyc inscribed 'common sense knowledge' as an arbitrarily extensible repository of propositions about the (one-world) world.⁴

Elish and boyd (2018) provide a concise critical history of the turn away from problem-solving and expert systems and towards the data-driven, statistical methods that comprise the currently dominant approaches of 'machine learning', 'neural networks', and their scaling up in convolutional neural networks or 'deep learning' systems.⁵ They trace how the turn to statistical methods was enabled by increases in computing power and a corporate embrace of Big Data beginning in the 1990s, followed by IBM's Watson project in the mid-2000s and the rebranding of Big Data as AI. Most recently, in response to growing evidence for the limits of data-driven approaches, critical practitioners within the field are calling for a return to symbolic logic as the basis for new 'hybrid' approaches (see Marcus, 2022; Heikkilä and Heaven, 2022). But this tacking back and forth between techniques fails to engage the starting premises and unexamined assumptions that critical genealogies of the field make evident (Dhaliwal et al., 2024).

AI as techniques and technologies

In service of demystification, the term 'AI' can be read as a label for currently dominant computational techniques and technologies that extract statistical correlations (designated as patterns) from large datasets, based on the adjustment of relevant parameters according to either internally or externally generated feedback. At the time of this writing, research and development under the sign of AI primarily comprise so-called machine learning and neural network approaches, applied to projects of natural language processing (NLP), the analysis or generation of various forms of 'content' (e.g. text, images, data sets and computer code) and automated decision/recommendation systems. A growing community of critical practitioners is providing clarifying explanations of the operations of these technologies, abstaining from anthropomorphism in favour of careful redescription. I offer just a few indicative examples here.

Pasquinelli (2019) identifies three components in the production of a machine learning system. The first involves the generation of 'training' data, corpora of digitized traces of activities or events 'captured' as images, text or numerical records. The second component is the algorithm designed to extract patterns from the training data, by constructing a complex statistical association between input and output, consisting of potentially billions of individually adjusted parameters. Finally, when the output produced by the statistical model shows an adequate alignment or 'fit' with the training data (as assessed by human operators), it can be applied to automate the classification of patterns or

predict the probability of the recurrence of a pattern in future data. Through their reliance on historical systems of classification and record-keeping, these techniques reproduce and amplify discriminatory practices. Perhaps most egregiously, they rely on the conflation of correlative and causal relations, a fallacy particularly problematic when it comes to prediction. As Pasquinelli (2019) emphasizes, this ‘is not a machine issue, but a political fallacy, when a statistical correlation between numbers within a dataset is received and accepted as causation among real entities in the world’.

In the field of NLP, Bender et al. (2021: 611) distinguish between language understanding and ‘string prediction tasks’ over massive training datasets. As they explain: ‘Contrary to how it may seem when we observe its output, an LM (language model) is a system for haphazardly stitching together sequences of linguistic forms it has observed in its vast training data, according to probabilistic information about how they combine, but without any reference to meaning: a stochastic parrot’ (2021: 616–17). They set out the costs (in CO₂ emissions, discriminatory content and exploited labour) and the unevenly distributed benefits of LMs. Demonstrating the capacities enabled by the scaling of parameters and datasets, these models have equally, the authors argue, revealed the limits of scale. They conclude with a call for ‘a re-alignment of research goals: Where much effort has been allocated to making models (and their training data) bigger and to achieving ever higher scores on leaderboards often featuring artificial tasks, we believe there is more to be gained by focusing on understanding how machines are achieving the tasks in question and how they will form part of socio-technical systems’ (2021: 618).

More generally, the quantification required to translate social practices into statistics includes processes of normalisation involved in data ‘reduction’, or the elimination of things that don’t fit, as well as by the information loss involved in rendering data into statistical distributions. As Broussard (2019: 103) emphasizes: ‘Data is made by people going around and counting things or made by sensors that are made by people. In every seemingly orderly column of numbers, there is noise. There is mess. There is incompleteness. This is life’. Yet dirty data confounds reliable computation; anomalies must be cleaned up to make functions run smoothly, and in that process, the irremediable contingency of signification disappears. As is now widely recognized among science and technology scholars, categorization is performative in that it works to write itself in and through the worlds that it orders.

AI as a floating signifier

Finally, AI can be defined as a sign invested with social, political and economic capital and with performative effects that serve the interests of those with stakes in the

field. Read as what anthropologist Claude Levi-Strauss (1987) named a floating signifier, ‘AI’ is a term that suggests a specific referent but works to escape definition in order to maximize its suggestive power. While interpretive flexibility is a feature of any technology, the thingness of AI works through a strategic vagueness that serves the interests of its promoters, as those who are uncertain about its referents (popular media commentators, policy makers and publics) are left to assume that others know what it is. This situation is exacerbated by the lures of anthropomorphism (for both developers and those encountering the technologies) and by the tendency towards circularity in standard definitions, for example, that AI is the field that aims to create computational systems capable of demonstrating human-like intelligence, or that machine learning is ‘a branch of artificial intelligence concerned with the construction of programs that learn from experience’ (Oxford Dictionary of Computer Science, cited in Broussard 2019: 91). Understood instead as a project in scaling up the classificatory regimes that enable datafication, both the signifier ‘AI’ and its associated technologies effect what philosopher of science Helen Verran has named a ‘hardening of the categories’ (Verran, 1998: 241), a fixing of the sign in place of attention to the fluidity of categorical reference and the situated practices of classification through which categories are put to work, for better and worse.

The stabilizing effects of critical discourse that fails to destabilize its object

Within science and technology studies, the practices of naturalization and decontextualization through which matters of fact are constituted have been extensively documented. The reiteration of AI as a self-evident or autonomous technology is such a work in progress. Key to the enactment of AI’s existence is an elision of the difference between speculative or even ‘experimental’ projects and technologies in widespread operation. Lists of references offered as evidence for AI systems in use frequently include research publications based on prototypes or media reports repeating the promissory narratives of technologies posited to be imminent if not yet operational. Noting this, Cummings (2021) underscores what she names a ‘fake-it-til-you-make-it’ culture pervasive among technology vendors and promoters. She argues that those asserting the efficacy of AI should be called to clarify the sense of the term and its differentiation from more longstanding techniques of statistical analysis and should be accountable to operational examples that go beyond field trials or discontinued experiments. !!!!!

In contrast, calls for regulation and/or guidelines in the service of more ‘human-centered’, trustworthy, ethical and responsible development and deployment of AI typically posit as their starting premise the growing presence, if not ubiquity, of AI in ‘our’ lives. Without locating

“AI ethics”

Bourdieu

invested actors and specifying relevant classes of technology, AI is invoked as a singular and autonomous agent outpacing the capacity of policy makers and the public to grasp 'its' implications. But reiterating the power of AI to further a call to respond contributes to the over-representation of AI's existence as an autonomous entity and unequivocal fact. Asserting AI's status as controversial, in other words, without challenging prevailing assumptions regarding its singular and autonomous nature, risks closing debate regarding its ontological status and the bases for its agency.

Troubling AI's uncontroversial reproduction

Recognizing the injurious consequences of AI rhetoric, on 8 March 2022, the Center on Privacy & Technology at Georgetown Law issued an announcement that began:

Words matter.

Starting today, the Privacy Center will stop using the terms 'artificial intelligence', 'AI', and 'machine learning' in our work to expose and mitigate the harms of digital technologies in the lives of individuals and communities (Tucker, 2022).

Avoiding references to AI or machine learning, Executive Director Emily Tucker writes, is 'a creative practice that we hope will support intellectual discipline'. She proposes that the term AI now stands in place of the 'scrupulous descriptions' that would aid public understanding of relevant technologies, as well as the corporate investments and data extractivism that are those technologies' conditions of possibility. 'To the extent that our words might make certain worlds even a little more or less possible for those to whom we speak and for whom we write', Tucker explains, 'we want to wield them carefully'.

As the editors of this special issue observe, the deliberate cultivation of AI as a controversial technoscientific project by the project's promoters pose fresh questions for controversy studies in STS (Marres et al., 2023). **I have argued here that interventions in the field of AI controversies that fail to question and destabilise the figure of AI risk enabling its uncontroversial reproduction. To reiterate, this does not deny the specific data and compute-intensive techniques and technologies that travel under the sign of AI but rather calls for a keener focus on their locations, politics, material-semiotic specificity and effects, including consequences of the ongoing enactment of AI as a singular and controversial object.** The current AI arms race is more symptomatic of the problems of late capitalism than promising of solutions to address them. Missing from much of even the most critical discussion of AI are some more basic questions: What is the problem for which these technologies are a solution? According to whom? How else could this problem be articulated, with what implications

for the direction of resources to address it? What are the costs of a data-driven approach, who bears them, and what lost opportunities are there as a consequence? And perhaps most importantly, how might algorithmic intensification be implicated not as a solution but as a contributing constituent of growing planetary problems – the climate crisis, food insecurity, forced migration, conflict and war, and inequality – and how are these concerns marginalized when the space of our resources and our attention is taken up with AI framed as an existential threat?⁶ These are the questions that are left off the table as long as the coherence, agency and inevitability of AI, however controversial, are left untroubled.

Acknowledgements

I am grateful to the editors of this special issue for their contributions to the sociology of technoscientific controversies that set the context for this essay and to the anonymous reviewers for their thoughtful comments and suggestions on how to strengthen and clarify the argument.

Declaration of conflicting interests

The author declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Funding

The author received no financial support for the research, authorship and/or publication of this article.

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Notes

1. Misplaced concreteness is drawn from Whitehead, 1948: 52; for discussion, see Haraway 1997: 146–147. Briefly, Whitehead is pointing to the attribution of intrinsic speaker and observer-independent qualities to the referents of abstract constructs, in ways that elide situated practices through which signs are made to attach to their meanings.
2. This is not to suggest an absence of critical engagement to date; see Raley and Rhee, 2023, Mackenzie, 2017.
3. For a rich set of resources that complicate and illuminate this ancestry, see *Histories of Artificial Intelligence: A Genealogy of Power* <https://www.ai.hps.cam.ac.uk/> (accessed July 2023).
4. For indicative critiques of the premise that there is one singular world, see Law 2015, de La Cadena and Blaser, 2018.
5. For an historically informed discussion of relations between ML, neural network and deep learning approaches, see Castelle, 2018.
6. Exemplified most recently in <https://www.safe.ai/statement-on-ai-risk> (accessed July 2023).

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